

# Flexible and Non-Flammable Zinc-Ion Battery Through Polymer Electrolytes

## Introduction

### Background

Batteries are a necessity to modern day society. From phones and laptops to cars, standardized lithium batteries have become increasingly more popular in many devices. However, in this project, we use zinc ion batteries which consists of an anode, cathode, and separator, and most use liquid electrolytes.

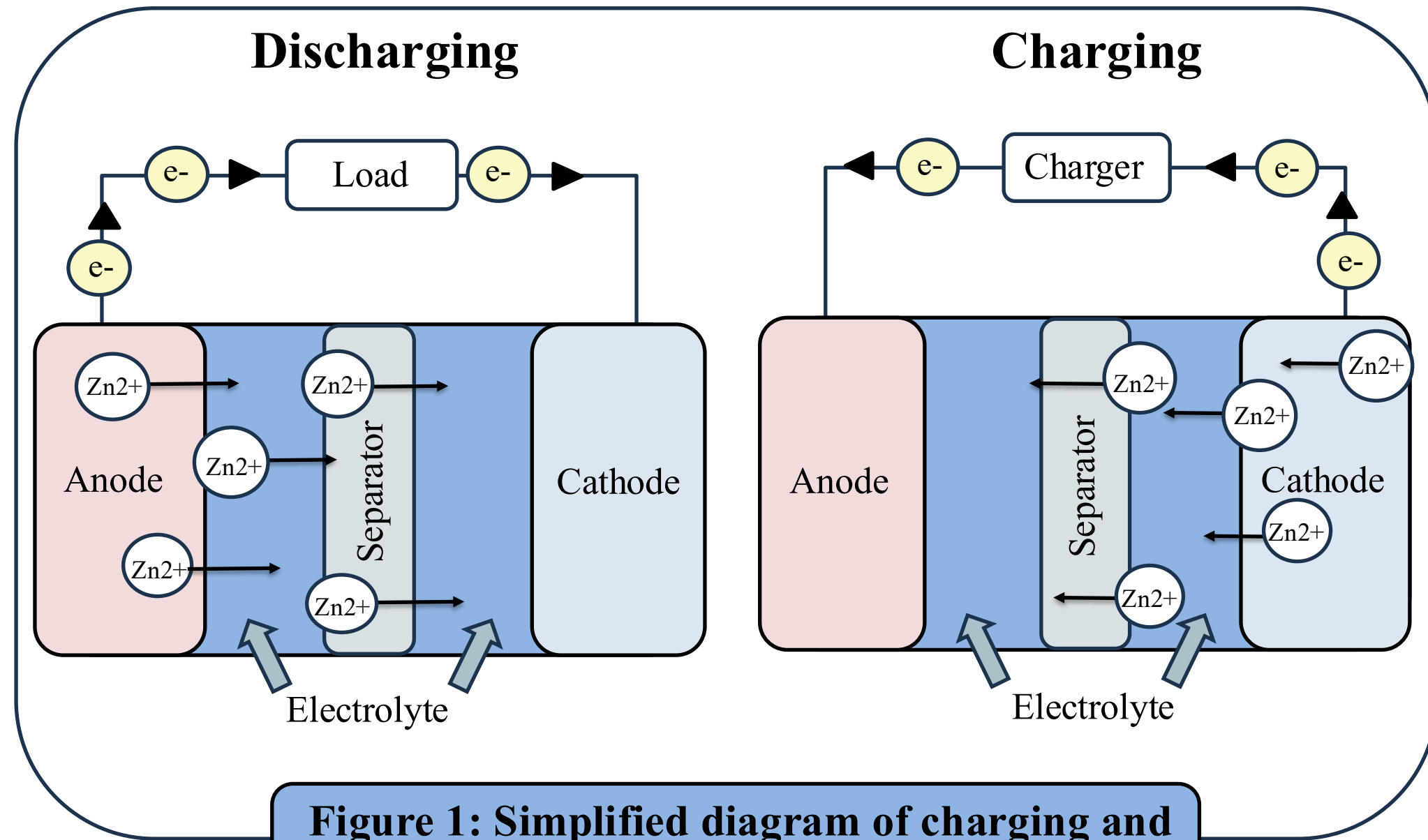


Figure 1: Simplified diagram of charging and discharging in a Zinc-ion battery

### Problem

Current standardized lithium-ion batteries with liquid electrolytes

- Not eco-friendly due to harm from extracting materials such as lithium and toxic leakage.
- Susceptible to severe dendritic growth and safety hazards.
- Resilience to external pressure and harsh environments.

### Effects

- Material extraction causes heavy water depletion, land degradation, and carbon dioxide emission.
- Toxic leakage is when poisonous electrolyte chemicals leak from batteries, causing health problems and pollution.
- Dendritic growth is when sharp pointy structures appear within the battery and pierce through the separator, causing short circuits, intense heat, and fires.
- Damaged batteries when exposed to harsh environments.

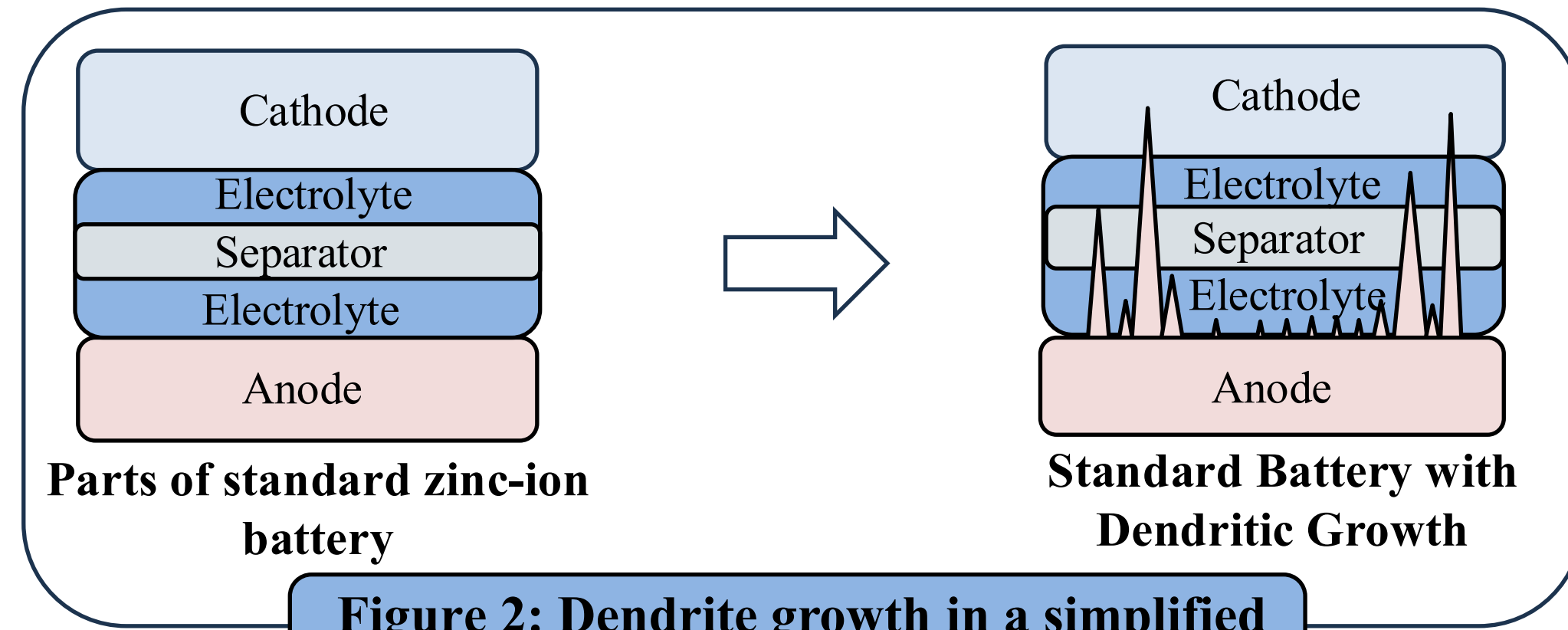


Figure 2: Dendrite growth in a simplified regular battery with liquid electrolytes

## Objectives

1. Substituting non-eco-friendly materials for another material such as zinc and reducing environmental harm.
2. Replacing liquid electrolyte chemicals with **gel-polymer electrolytes** to avoid problems listed above and to make the battery safer.
3. Comparing **electrochemical efficiency, retention, performance** between standard zinc-ion liquid electrolyte batteries and the new gel-polymer electrolyte batteries we create.
4. Improving current batteries by **improving resilience, stability, and functionality**.

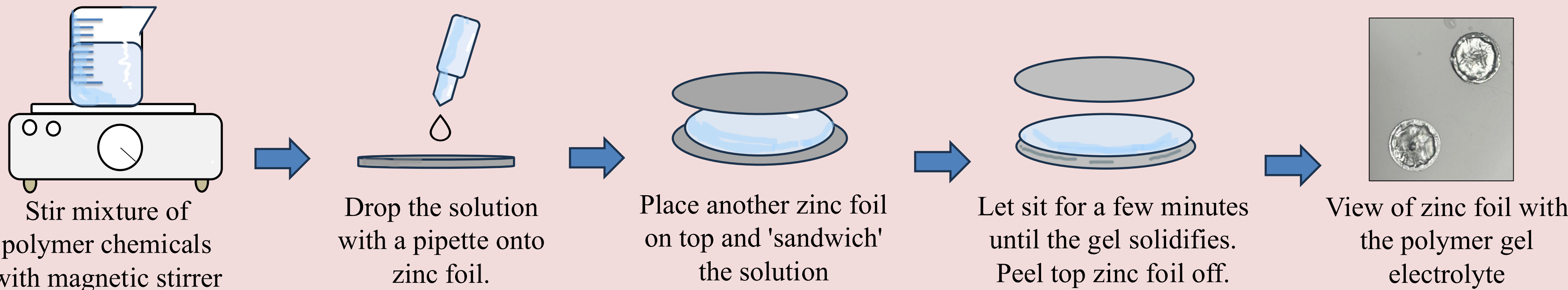
## Methodology

### Chemical Recipe for Liquid and PAM Hydrogel-Polymer Electrolytes in Zinc Batteries

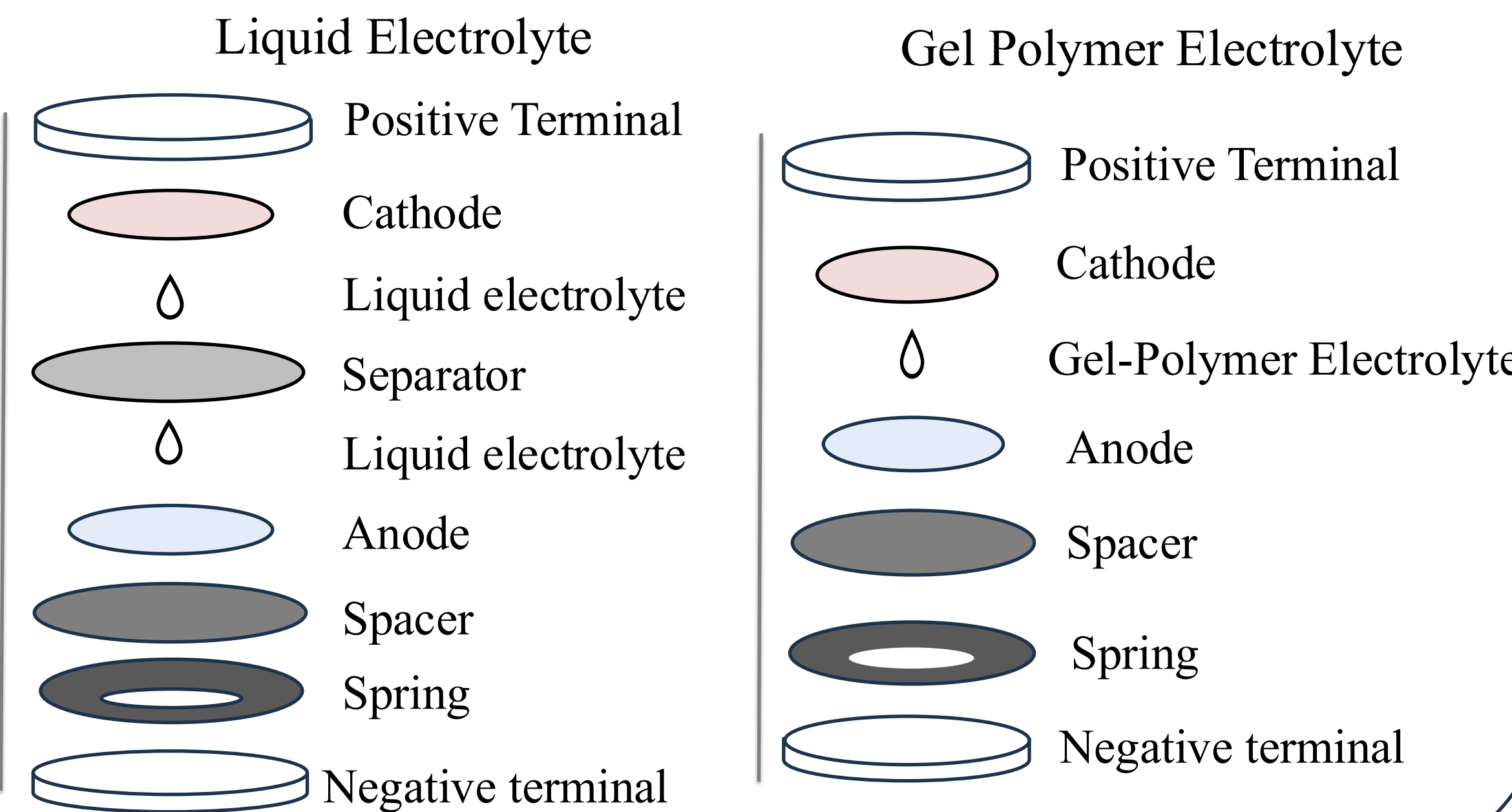
**PAM Hydrogel Electrolyte Recipe:** 1g Acrylamide, 0.04g N,N'-methylenebisacrylamide monomers, 0.08 potassium persulfate, dissolve in 10 mL of 2M trifluoromethanesulfonate solution (fabrication process below)

**Zinc Liquid Electrolyte Recipe:** Zinc Salt, Water (mixed together)

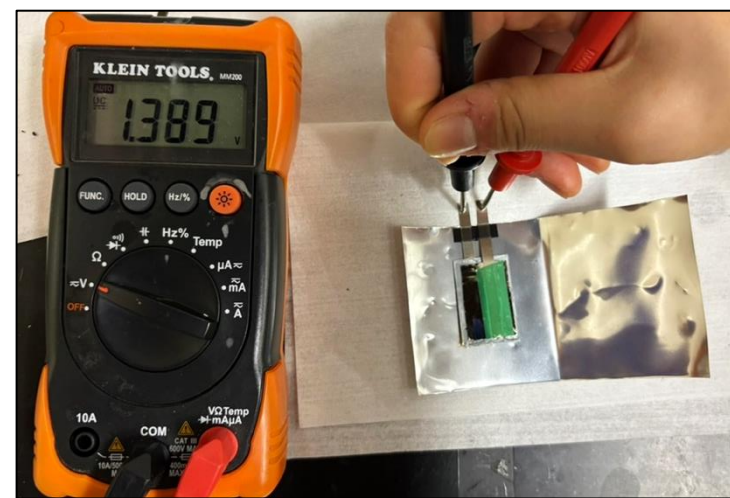
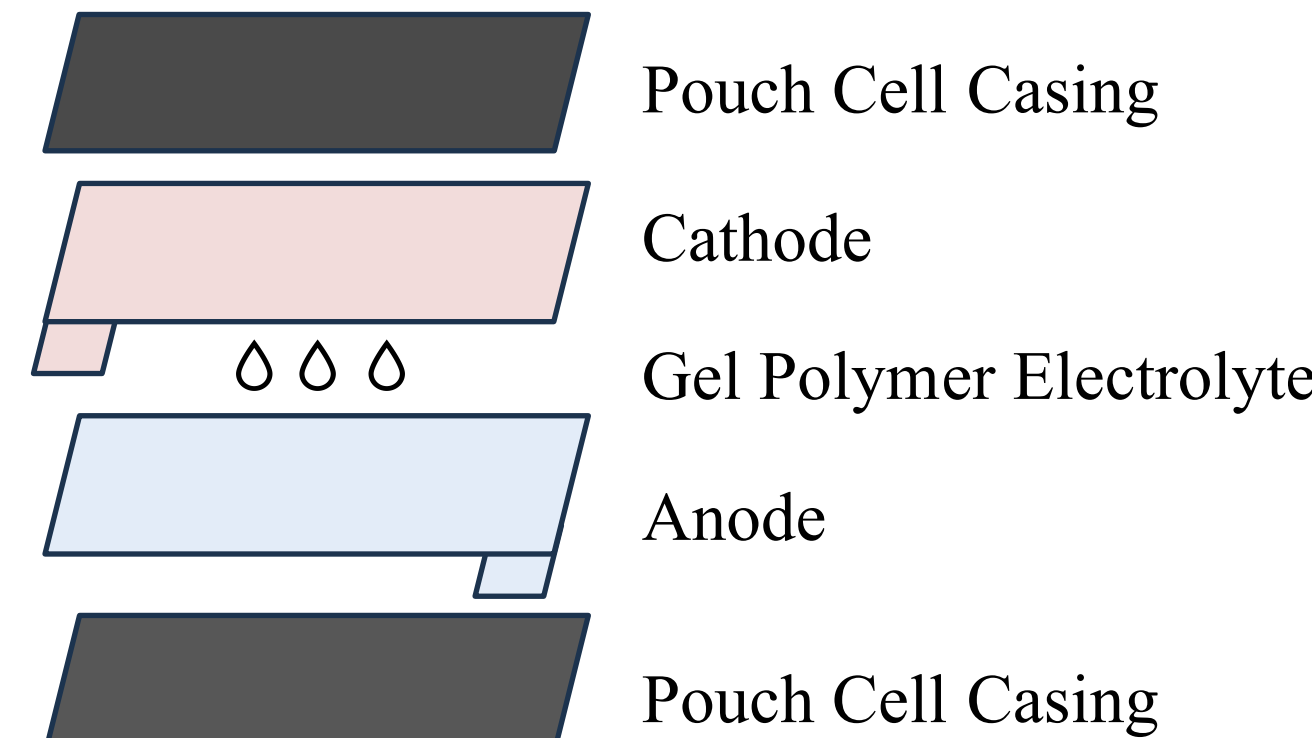
### PAM Hydrogel-Polymer Electrolyte Fabrication for Zinc Batteries



### Zinc-Ion Coin Cell



### Zinc-Ion Pouch Cell



Unsealed Zinc-Ion Cell Pouch with a Multimeter measuring voltage

## Results

### Cycle Retention (Liquid vs. Polymer Electrolyte)

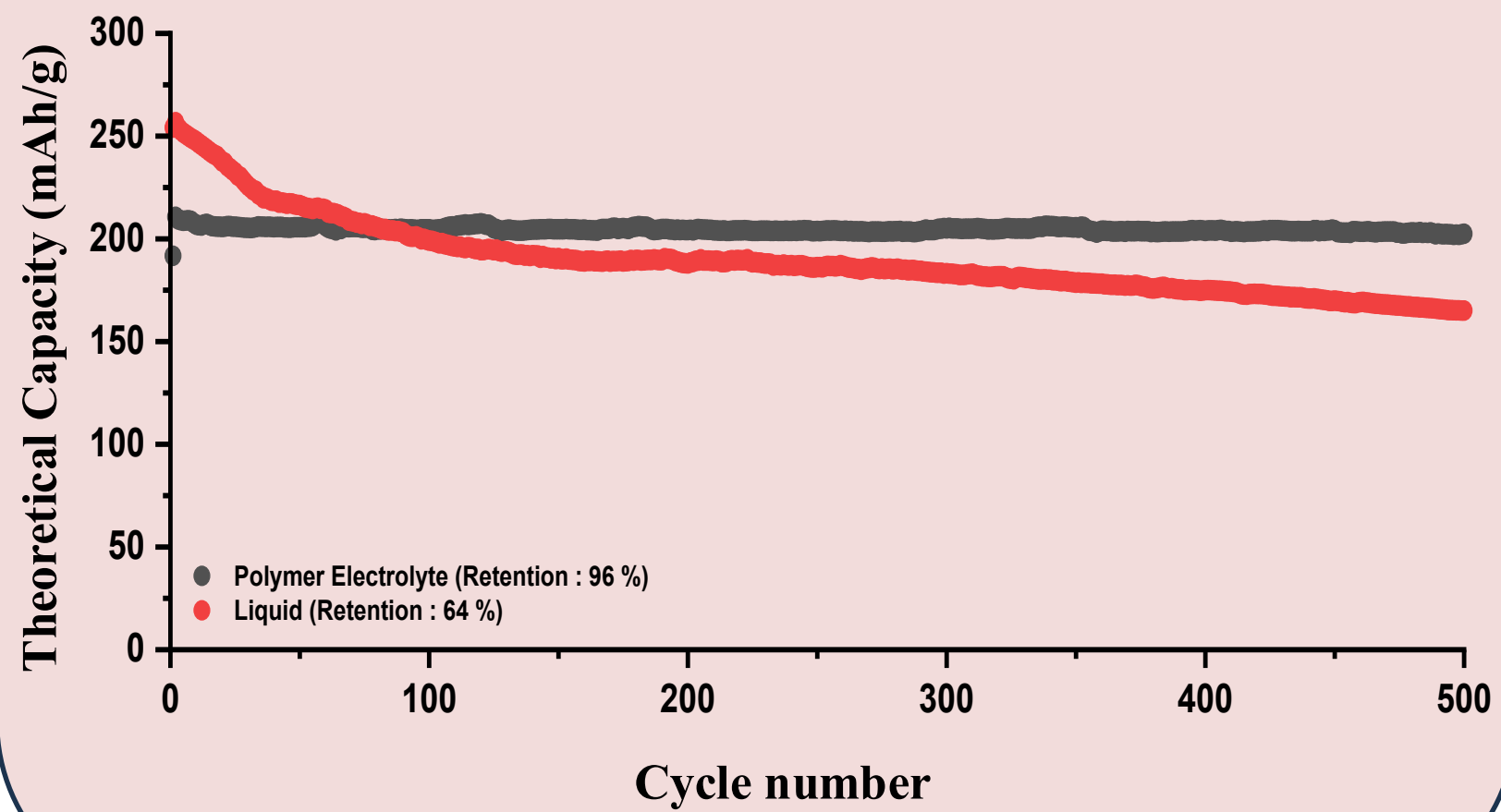


Figure 3: Theoretical capacity over 500 cycles comparing liquid and polymer electrolytes

### CV of Zinc-Ion Battery with Polymer Electrolyte

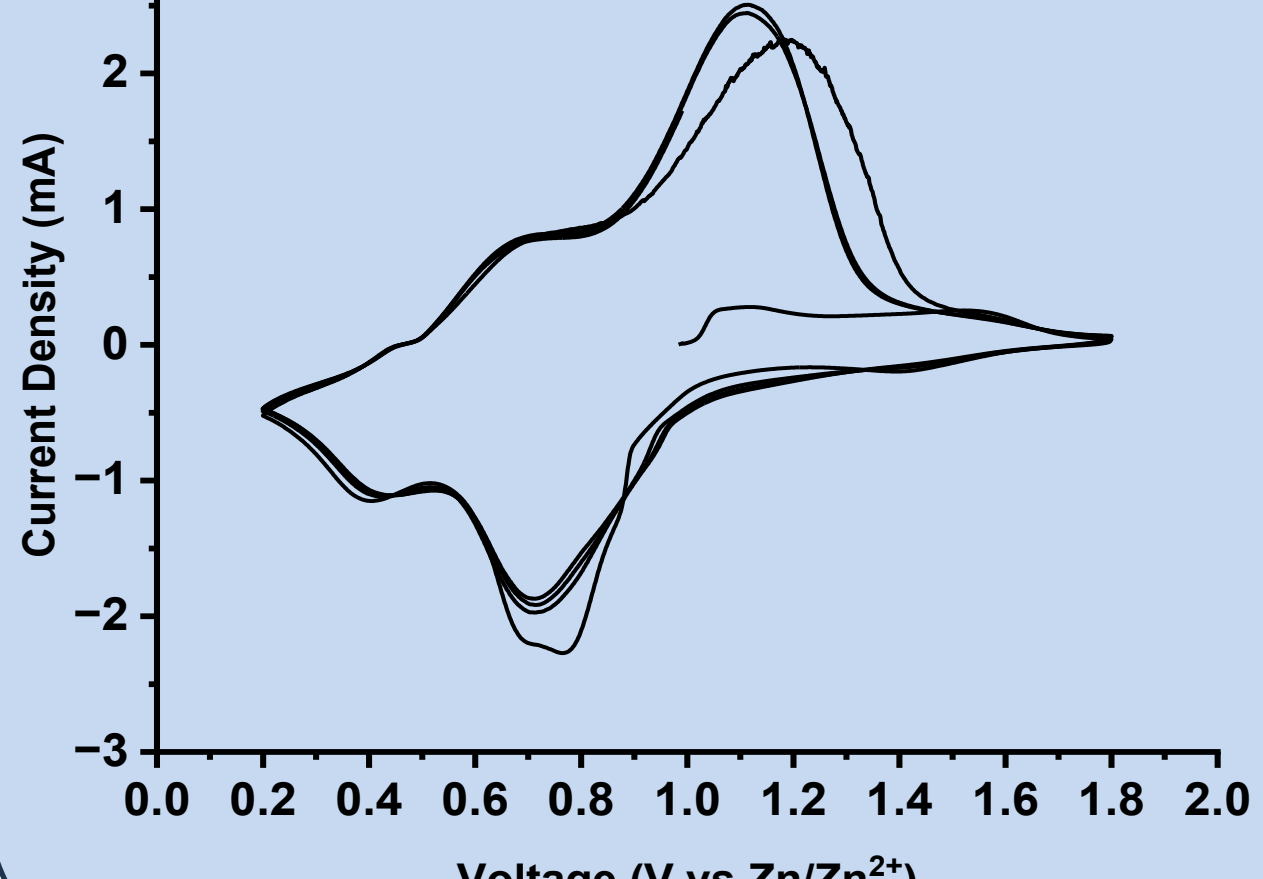


Figure 4: Cyclic Voltammetry demonstrating electrochemical support of ion transportation

### What this means:

Comparison between the standardized liquid electrolyte and the polymer electrolyte in **2 aspects**  
**Cycle Retention:** Our polymer electrolyte yields better results, depicting less capacity loss through cycles. Values yield 96% and 64% respectively (Figure 3).  
**Cyclic Voltammetry:** Our polymer electrolyte facilitates ion transport, and there are two clear peaks representing redox and oxidation. The polymer electrolyte has good electrochemical based off our data (Figure 4).  
**TGA Test:** The polymer electrolyte has significantly better thermal stability, as the polymer itself decomposes at 466 °C, seen in the second dip in Figure 5. However, standard liquid electrolytes have poor thermal stability, as they're made of zinc-salt dissolved in water and decomposes at around 100 °C due to water's boiling point. At high temperatures, the polymer electrolyte proves to be more advantageous (Figure 5).  
**Electrolyte Resistance:** Our polymer electrolyte has lower resistance, allowing better ionic conductivity. Values yield 17.4 and 25.7 ohms respectively (Figure 6).

### PAM Hydrogel TGA test

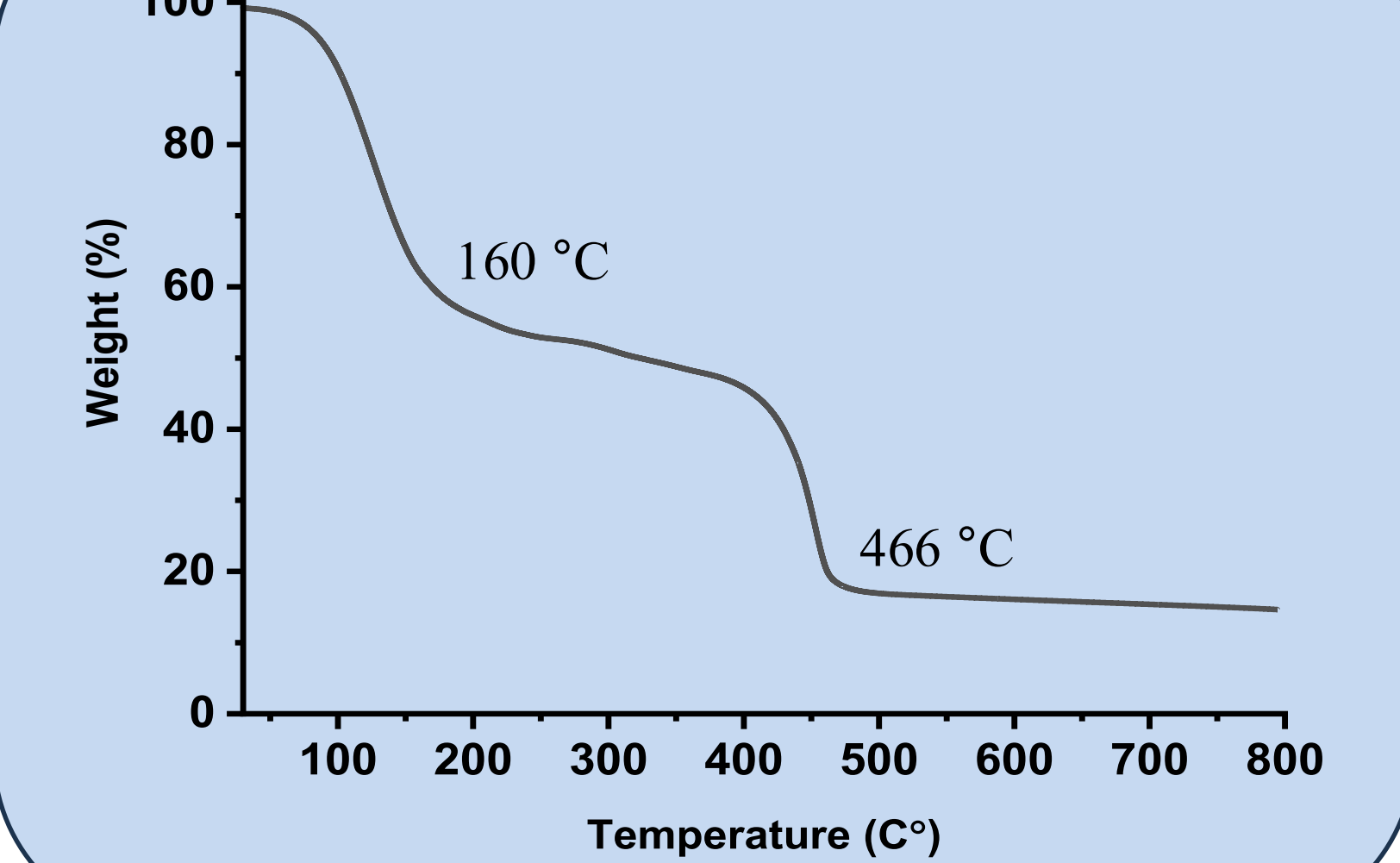


Figure 5: Thermogravimetric analysis of weight loss at 2 peaks for PAM hydrogel

### EIS Nyquist Plot

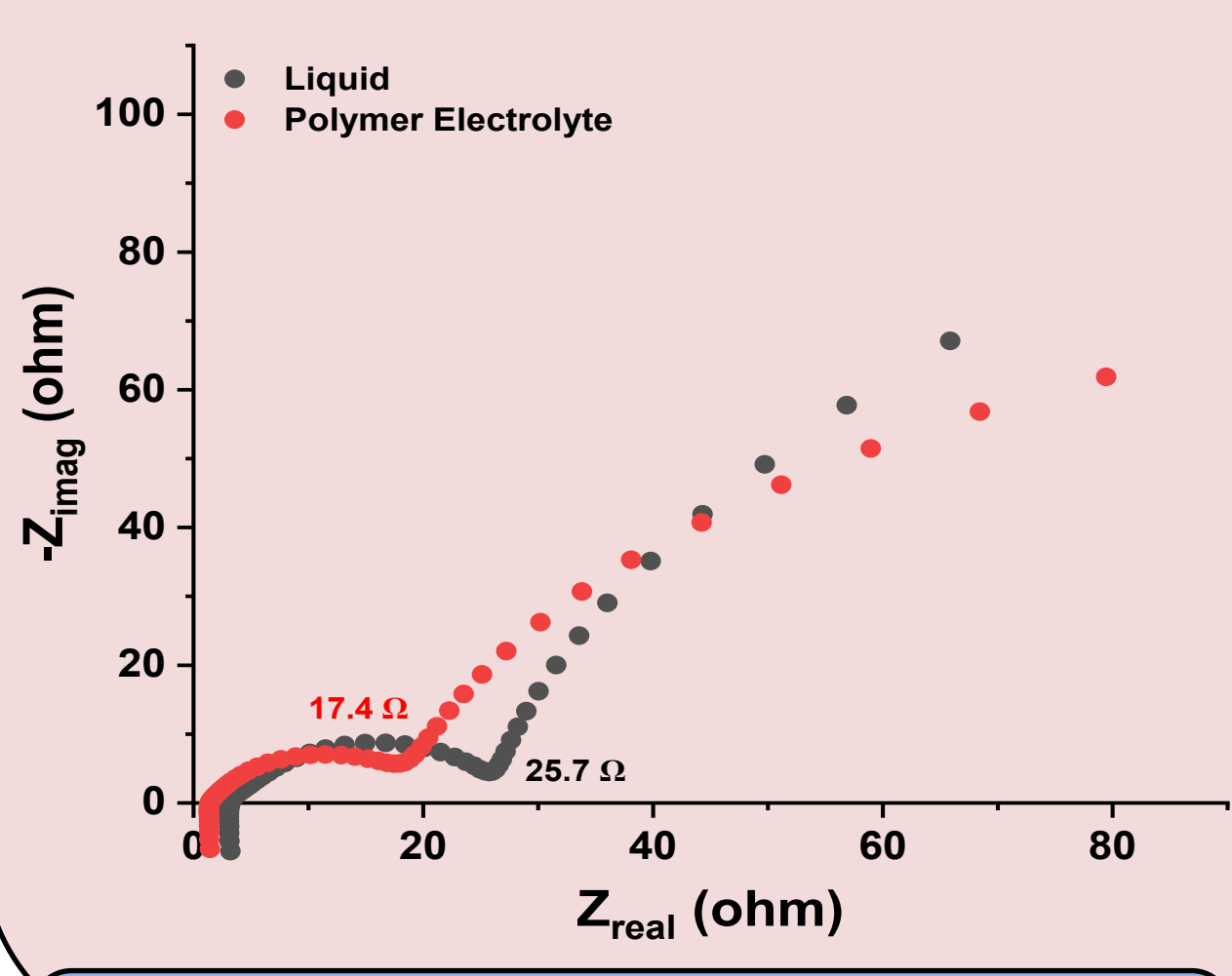


Figure 6: Nyquist Plot measuring resistance of liquid and polymer electrolytes

## Conclusion and Applications

Through our approaches, we are able to provide for solutions to our problems while providing even more advantages and meet our objectives.

1. We substitute non-eco-friendly materials such as lithium for zinc, which is a lot more common and better for the environment. Zinc is also cost effective and non-toxic, and it has a high capacity retention.
2. We replace liquid electrolytes for PAM hydrogel-polymer electrolytes to prevent toxic leakage and dendritic growth. The polymer electrolyte is also non-flammable, making the likelihood of a fire much smaller.
3. We run tests to find the performance of the polymer electrolyte in comparison to the liquid electrolyte by using zinc-coin cells with figures 3-6.
4. We create more resilient batteries with the usage of the pouch cell and the PAM polymer electrolyte. The zinc-ion pouch cell allows for flexibility, allowing for new applications and resilience against high stress environments. The polymer electrolyte itself has more thermal stability, allowing for usage under higher temperature environments. The battery can also withstand a certain amount of damage and punctures.

### Potential Applications

#### 1. Medical applications

The zinc-ion polymer electrolyte batteries we made are bio-compatible, and since they're flexible, they could be used for medical sensors, implants, and pumps. Flexible bio-compatible batteries could be revolutionary for real-time health monitoring devices (Chen et al., 2025).

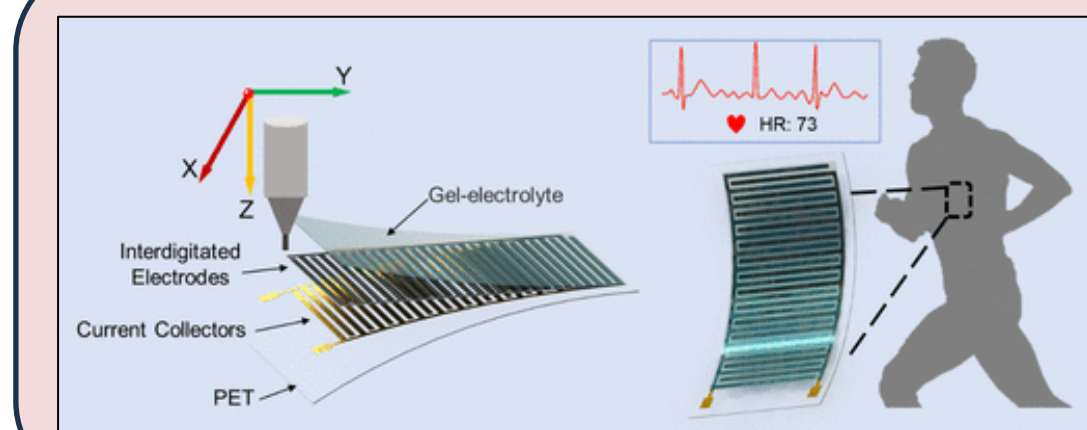


Figure 7: Diagram from *ACS Appl. Mater. Interfaces* 2025, 17, 16, 23860-23871. Image demonstrates medical sensor prototype with a flexible battery.

#### 3. Energy storage systems

The zinc-ion polymer electrolyte batteries can be safe, reliable energy systems due to their high capacity retention and long battery life.

#### 4. Devices in high stress environments

Due to the polymer electrolyte battery's flexibility, there is mechanical resilience. The PAM hydrogel polymer electrolyte also offers chemical resilience, as it's shown to withstand temperatures to around 466 °C, while most liquid electrolytes can only withstand temperatures of around 100 °C. The qualities mentioned allows for use in harsh environments.

## Bibliography

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